

Group-Aggregate Features Defeat Spatial Cross-Validation: A Case Study in Urban Storefront Turnover

Bahodir Nazarov*

September 19, 2026

Abstract

Spatially blocked cross-validation is the standard defence against optimistic performance estimates on geographic data. We show that it fails silently when block size is not chosen with respect to the spatial support of the features, and that the dominant leakage channel is not fine-grained spatial autocorrelation but *group-aggregate features*—leave-one-out target encodings over administrative units.

On a task built from 1.21 million City of Chicago business licences spanning 1995–2026, we predict which of 135,553 storefronts exhibit excess tenant turnover. Random k -fold cross-validation reports ROC AUC 0.970 and average precision 0.756 ($15.1\times$ over the base rate). Spatially blocked cross-validation over the same data and model reports 0.693 and 0.114 ($2.3\times$)—an average-precision inflation of more than sixfold.

Three findings follow. First, blocking at 100 m–1 km is statistically indistinguishable from no blocking at all (AUC 0.971–0.977); performance collapses only once blocks approach administrative scale. Practitioners who adopt spatial cross-validation but choose small blocks obtain inflated numbers carrying a methodologically respectable label. Second, leakage is concentrated in model families by flexibility: logistic regression inflates by 0.013 AUC while gradient boosting inflates by 0.277. Under honest evaluation all families are equivalent (0.62–0.70), so random splits corrupt model *selection*, not merely model assessment. Third, ablation localises the leak: ward and community-area aggregates inflate by 0.321 AUC against 0.034 for 50–500 m neighbourhood features. Because a leave-one-out group mean is near-constant within its group, it acts as a group identifier, and the evaluation split must align with the grouping used to construct it.

The substantive result survives honest evaluation: excess turnover concentrates at individual addresses rather than districts, with neighbours only weakly elevated and tenancies failing roughly $1.5\times$ faster per renewal cycle. All inputs are public domain and the pipeline is released in full.

1 Introduction

Applied machine learning on geographic data has a well-documented evaluation problem. Observations close in space share unobserved causes, so a randomly partitioned test set is not independent of the training set, and performance estimates are optimistic. The remedy—partitioning by spatial block rather than at random—is established in ecology and remote sensing [Roberts et al., 2017, Ploton et al., 2020, Valavi et al., 2019].

This paper reports that the remedy, as commonly applied, is easy to administer incorrectly in a way that produces no visible symptom. We make three claims.

*Correspondence: b.nazarov0309@gmail.com. Code and derived data: <https://github.com/naazzarov/storefront-turnover-leakage>

Blocking below the feature support is inert. Spatial blocking is typically described as a binary choice: random splits are wrong, spatial splits are right. We find it is a continuum governed by a specific quantity—the spatial support of the features. Our features summarise neighbourhoods out to 500 m. Blocks of 100 m, 250 m, 500 m and 1 km yield AUC 0.977, 0.977, 0.976 and 0.971, against 0.970 for random splits. A practitioner who reads the literature, adopts spatial cross-validation, and selects a 500 m block obtains a fully inflated estimate while believing the problem addressed. This is a worse position than using random splits knowingly.

The dominant channel is group aggregation, not proximity. We expected neighbourhood features to carry the leak. Ablation shows otherwise: ward and community-area aggregates inflate AUC by 0.321, an order of magnitude more than the 50–500 m neighbourhood features at 0.034. The mechanism is not spatial in any essential way. A leave-one-out group mean varies little within its group, so it encodes group membership, and under a random split the model recovers each group’s label rate from training rows sharing that group. This is target encoding [Micci-Barreca, 2001], and leave-one-out construction—the standard precaution—does not prevent it. The requirement is that the evaluation partition align with the grouping used for the encoding. The finding therefore applies to any grouped data with target-encoded features, not only to geography.

Leakage is a function of model capacity. Inflation ranges from 0.013 AUC for logistic regression to 0.277 for gradient boosting. Under blocked evaluation every family lands between 0.62 and 0.70. Random splits thus misrank models: a practitioner comparing families under random cross-validation concludes that gradient boosting substantially outperforms logistic regression, when honestly evaluated they are equivalent. The consequence is a corrupted model-selection decision, which survives into deployment in a way a single inflated number does not.

We develop these claims on a concrete task with an independently interesting answer: identifying commercial addresses that turn over tenants far more often than their context predicts.

2 Related work

Spatial cross-validation. Roberts et al. [2017] give the standard treatment of blocked designs for structured dependence, and Ploton et al. [2020] show that ignoring spatial autocorrelation inflates reported accuracy in ecological models. Valavi et al. [2019] provide tooling. This literature establishes *that* blocking is needed. It gives comparatively little guidance on *how large* blocks must be relative to the features, which is the quantity we find decisive, and it is framed around species distribution rather than the administrative-unit aggregates that dominate urban prediction.

Leakage. Kaufman et al. [2012] give the canonical taxonomy of leakage; Kapoor and Narayanan [2023] document its prevalence across published work in many fields. Target encoding and its leakage risks are discussed by Micci-Barreca [2001] and Prokhorenkova et al. [2018], generally in the context of a single train/test split rather than the interaction between encoding groups and evaluation partitions that we isolate here.

Business survival. Firm exit has a long empirical literature [Dunne et al., 1988], typically at firm or sector level. Work on location-level persistence is comparatively sparse, and municipal licence registers remain an under-used source despite recording entry and exit dates directly.

3 Data and task

3.1 Source

The City of Chicago publishes every business licence issued since 1995 as public domain data.¹ We use the complete extract of 1,207,055 licence records retrieved 2026-09-18.

We note that the portal’s one-click CSV export returns approximately 228k rows with HTTP 200 and no warning. All results here use the paginated API with the row count verified against the server’s own total. We mention this because a truncated extract produces plausible output and is difficult to detect downstream.

3.2 From licences to tenancies

A licence is not a storefront, and three properties of the register would inflate apparent turnover if taken at face value. Licences renew on one- or two-year terms, so a continuously trading business appears as many rows. Businesses hold several licence classes concurrently (a restaurant may hold food, liquor and sidewalk-cafe licences). Ownership restructuring or renaming ends one licence and begins another while trading is uninterrupted.

We therefore collapse licences into tenancies in two stages. Licences sharing an account number at one address are merged into a single occupancy when consecutive spells are separated by less than 200 days, absorbing renewals and concurrent classes. Occupancies by different account holders at one address are then merged when separated by less than 30 days, on the grounds that a handover with no vacancy is more likely a restructuring than a failure and replacement. The succession threshold is deliberately short, since aggressive merging suppresses the signal being measured.

Addresses are normalised before grouping (suffix and directional canonicalisation, ordinal collapse, accent stripping), with unit designators retained so that separate suites in one building remain separate storefronts.

3.3 Censoring and quantization

Two measurement properties constrain what may be inferred.

Licence terms extend into the future—this extract contains expiry dates as late as 2030. Anchoring the observation point to the latest date present would place it years after the data ends, and every currently trading business would appear as a completed tenancy credited with unelapsed time. We anchor to the latest issue date (2026-09-18); 38,143 of 194,203 tenancies are right-censored.

Tenancy duration is quantized to licence terms. Only 6.2% of tenancies end with a recorded cancellation and therefore an exact date; the remaining 93.8% lapse, and are known only to have ended within a term. Observed durations concentrate on exact multiples of licence terms (729–730 days, 364–365, 1460). Duration in days is consequently closer to a statement about the municipal renewal calendar than about trading life, and we use renewal cycles as the time axis for survival analysis.

3.4 Target definition

Raw tenant count conflates three things that are not properties of the address: exposure (an address first licensed in 1995 has had thirty years to turn over), sector base rate (hospitality fails more often than professional services), and district (some corridors churn throughout).

¹Dataset `r5kz-chrr`, <https://data.cityofchicago.org>.

We therefore fit expected turnover with a negative-binomial GLM using $\log(\text{exposure})$ as offset and dominant licence class and community area as covariates, and define *excess turnover* as the residual. Locations in the upper 5% of excess are labelled positive. A location is not positive for turning over often, but for turning over more than its own context predicts.

Restricting to locations with at least eight years of exposure leaves 135,553 locations across 78 community areas, of which 6,778 are positive. Positive locations average 3.12 tenants against 1.13 for the remainder.

We note that community area enters the target definition. Comparisons of positives against negatives at community-area level are therefore partly circular, and we exclude them from substantive claims; the 50–500 m and building-level comparisons are out-of-model and are used instead.

3.5 Features

All contextual features are computed leave-one-out, excluding the focal location. A neighbourhood statistic that includes the focal location partially reproduces the label.

Features comprise: leave-one-out mean turnover among neighbours within 50, 200 and 500 m and neighbour counts at each radius; leave-one-out means over ward and community area; building-level context (units in building, leave-one-out turnover among other units); and coordinates. Exposure terms are excluded from predictive experiments because exposure enters the target definition.

4 Experimental protocol

We compare partitioning schemes holding data, features and model fixed.

Random. Stratified 5-fold, shuffled—the default in common tooling.

Spatial grid. Space is tiled into squares of edge $s \in \{100, 250, 500, 1000, 2000, 5000\}$ m; whole squares are assigned to folds, so locations within a square cannot straddle the split.

Administrative. Whole community areas (78) are assigned to folds, balanced greedily by size.

We report ROC AUC (mean and standard deviation across folds) and average precision computed on pooled out-of-fold predictions. Average precision is reported because the positive rate is 5%, at which AUC is a flattering summary; the base rate gives a directly interpretable reference.

For the substantive spatial contrast we use a permutation null that shuffles labels *within* community areas, preserving each district’s turnover level so that only the assignment of labels inside it is randomised. A citywide shuffle would destroy district-level clustering and render almost any contrast significant.

5 Results

5.1 Block size must exceed feature support

Table 1 and Figure 1 show performance is flat from random splits through 1 km blocks and falls only beyond 2 km. Since features summarise neighbourhoods to 500 m, blocks at or below that scale cannot sever the dependence they induce: information propagates as far as the blocks are wide.

The practical implication is a diagnostic rather than a fixed recommendation. Sweeping block size and observing a flat curve indicates blocks are too small relative to feature support; the curve’s decline locates the scale at which dependence is actually broken.

Partition	Blocks	ROC AUC	AP	Lift
Random (none)	—	0.970 ± 0.004	0.756	$15.1\times$
Grid 100 m	20,822	0.977 ± 0.003	0.755	$15.1\times$
Grid 250 m	7,111	0.977 ± 0.002	0.748	$15.0\times$
Grid 500 m	2,202	0.976 ± 0.002	0.760	$15.2\times$
Grid 1 km	632	0.971 ± 0.005	0.731	$14.6\times$
Grid 2 km	189	0.942 ± 0.014	0.618	$12.4\times$
Grid 5 km	40	0.825 ± 0.091	0.208	$4.2\times$
Community area	78	0.693 ± 0.062	0.114	$2.3\times$

Table 1: Performance against partitioning scheme. Data, features and model are identical throughout; only the partition changes. Blocking at 100 m–1 km is indistinguishable from no blocking.

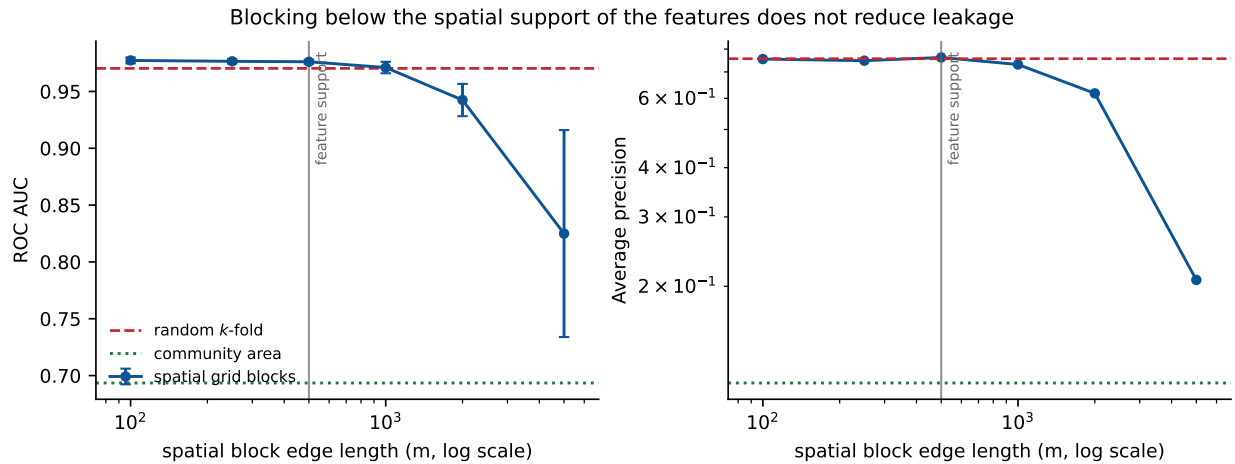


Figure 1: Performance against spatial block size. The curve is flat until blocks exceed the 500 m spatial support of the features (grey line), then falls. A practitioner selecting a block below that scale obtains an estimate indistinguishable from random splits.

5.2 Leakage scales with model capacity

Table 2 and Figure 2 show inflation rising with model flexibility. The consequence extends beyond reporting. Under random cross-validation gradient boosting appears to outperform logistic regression by 0.261 AUC; under blocked evaluation the gap is -0.003 . Random splits therefore do not merely inflate a figure, they invert a model-selection decision, and the inflation is largest precisely for the model families practitioners most often prefer.

5.3 Group aggregates dominate the leak

Table 3 and Figure 3 localise the leak to administrative aggregates. Four features alone reach AUC 0.978 under random splits and 0.657 under blocked splits. Proximity features—the expected culprit—contribute little.

The mechanism is group identity rather than distance. A leave-one-out mean over a ward is near-constant within that ward, so it serves as a ward identifier; under a random split, training rows disclose the ward’s label rate to test rows in the same ward. Leave-one-out construction

Model	Random AUC	Blocked AUC	Δ	Random AP	Blocked AP
Logistic regression	0.709	0.696	+0.013	0.098	0.098
k NN ($k = 25$)	0.686	0.616	+0.070	0.092	0.072
Random forest	0.933	0.686	+0.247	0.670	0.086
Gradient boosting	0.970	0.693	+0.277	0.756	0.114

Table 2: Inflation by model family. Under blocked evaluation all families are equivalent; under random splits they are not.

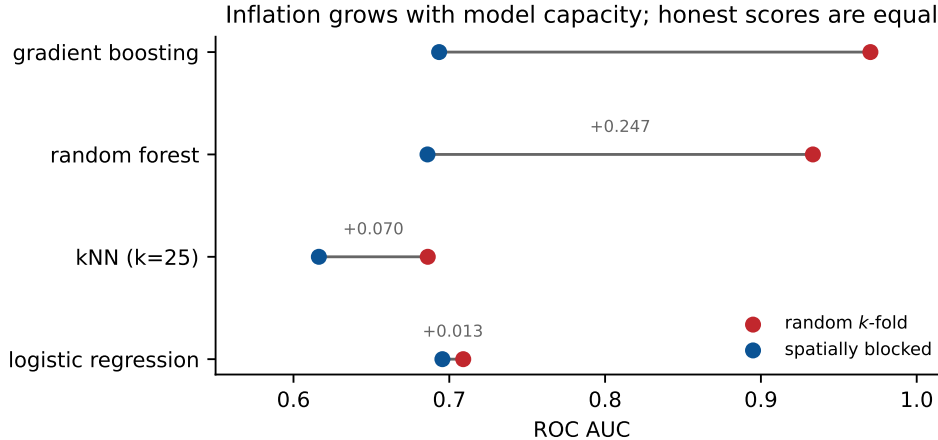


Figure 2: Paired random and blocked scores by model family. Honest scores coincide; random-split scores do not, so the ranking is an artefact of the partition.

removes the focal observation but not the remaining $n - 1$, and with large groups this is negligible protection.

This reframes the requirement. What must align is not blocks with geography but *the evaluation partition with the grouping used to construct aggregate features*. Where those groups are administrative units, spatial blocks of arbitrary size do not satisfy it. Where the groups are not spatial at all—user, merchant, session—the same failure arises with no spatial character, and the same remedy applies.

5.4 The substantive finding

Under honest evaluation the urban result stands. Excess turnover concentrates at individual addresses rather than districts. Leave-one-out neighbour turnover exceeds the within-area permutation null by +0.104 tenants at 50 m ($p = 0.001$, null $+0.034 \pm 0.003$), +0.068 at 200 m and +0.040 at 500 m, against a focal difference of +1.99 tenants. Neighbours of high-turnover addresses are elevated by roughly 5%; the addresses themselves by roughly 176%.

Discrete-time survival over renewal cycles (Figure 5) shows tenancies at these locations ending faster at every stage: first-cycle hazard 0.337 against 0.233, and survival to the eighth cycle of 0.119 against 0.278. The hazard ratio is approximately constant across cycles, consistent with proportional hazards.

Context predicts the label only weakly (AUC 0.693, AP $2.3 \times$ base rate). Most excess turnover is therefore not explained by observable surroundings.

Feature family	k	Random	Blocked	Δ
Area aggregates (ward, community area)	4	0.978	0.657	+0.321
Raw coordinates	2	0.659	0.595	+0.064
Neighbourhood LOO (50–500 m)	6	0.722	0.687	+0.034
Density counts	3	0.694	0.672	+0.021
Building context	2	0.508	0.506	+0.001

Table 3: Inflation by feature family, gradient boosting. Administrative aggregates inflate an order of magnitude more than proximity features.

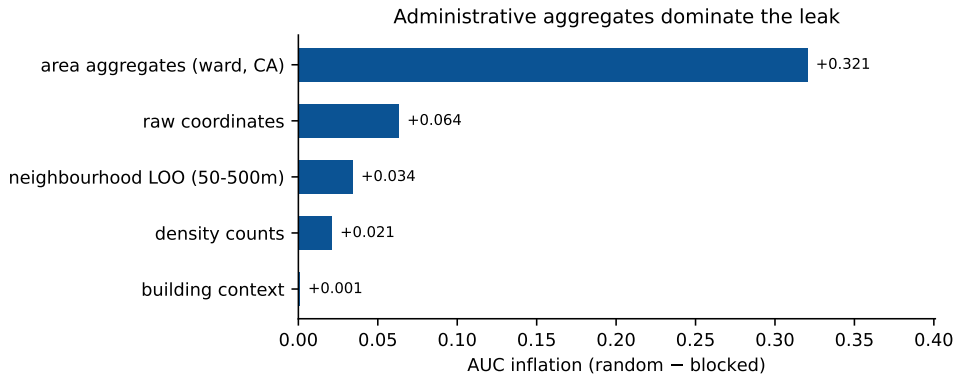


Figure 3: AUC inflation by feature family. Administrative aggregates dominate; proximity features contribute little.

5.5 Sensitivity to the target definition

Table 4 reports agreement between the three definitions. The exposure-adjusted definitions overlap substantially (Jaccard 0.632), but the unadjusted count-based definition overlaps both only weakly (0.187, 0.193). This is a genuine limitation and we state it plainly: which addresses are called high-turnover depends materially on whether exposure and sector are adjusted for, and a reader who prefers a raw-count definition would be studying a substantially different set of locations.

We note that this sensitivity concerns the substantive finding rather than the evaluation findings. The leakage results compare partitioning schemes holding the target fixed, so they are unaffected by which definition is chosen; the mechanism—group aggregates encoding group identity—depends on the feature construction, not on the label.

6 Recommendations

1. **Partition by the grouping used for aggregate features.** If features include group-level statistics, folds must respect those groups. This subsumes spatial blocking when groups are geographic and applies unchanged when they are not.
2. **Report a block-size sweep.** A flat performance curve across block sizes indicates blocks below feature support. The curve costs little and makes the claim auditable.
3. **Treat leave-one-out as insufficient.** It removes the focal observation only. With large groups the remaining members carry essentially the full signal.

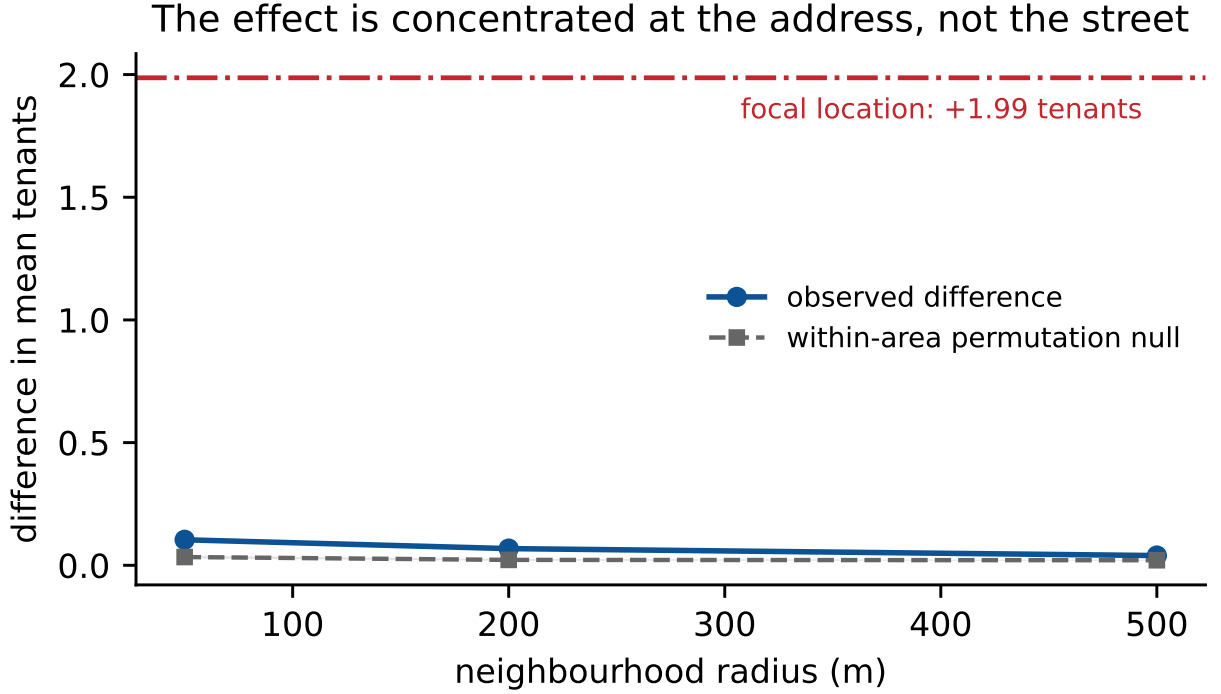


Figure 4: Difference in mean tenant count between high-turnover locations and the remainder, by neighbourhood radius, against a permutation null that shuffles labels within community areas. The focal difference exceeds the neighbour difference by an order of magnitude.

	Count	Rate	Excess
Count (≥ 4 tenants)	1.000	0.187	0.193
Rate (tenants per decade, top 5%)	0.187	1.000	0.632
Excess (model residual, top 5%)	0.193	0.632	1.000

Table 4: Jaccard overlap between the sets of locations flagged by three target definitions.

4. **Compare model families under the final protocol.** Capacity-dependent inflation means selection under random splits can invert under blocked splits.
5. **Report average precision with the base rate** for imbalanced targets.

7 Limitations

Licence records proxy trading. A lapse may reflect administrative change rather than closure; our succession merge is a conservative but imperfect correction, and residual misclassification inflates apparent turnover.

Duration is interval-censored and quantized, which is why we use renewal cycles; absolute tenure figures should not be read from this data.

Community area enters both the target definition and the partitioning, so area-level comparisons are circular and excluded from substantive claims.

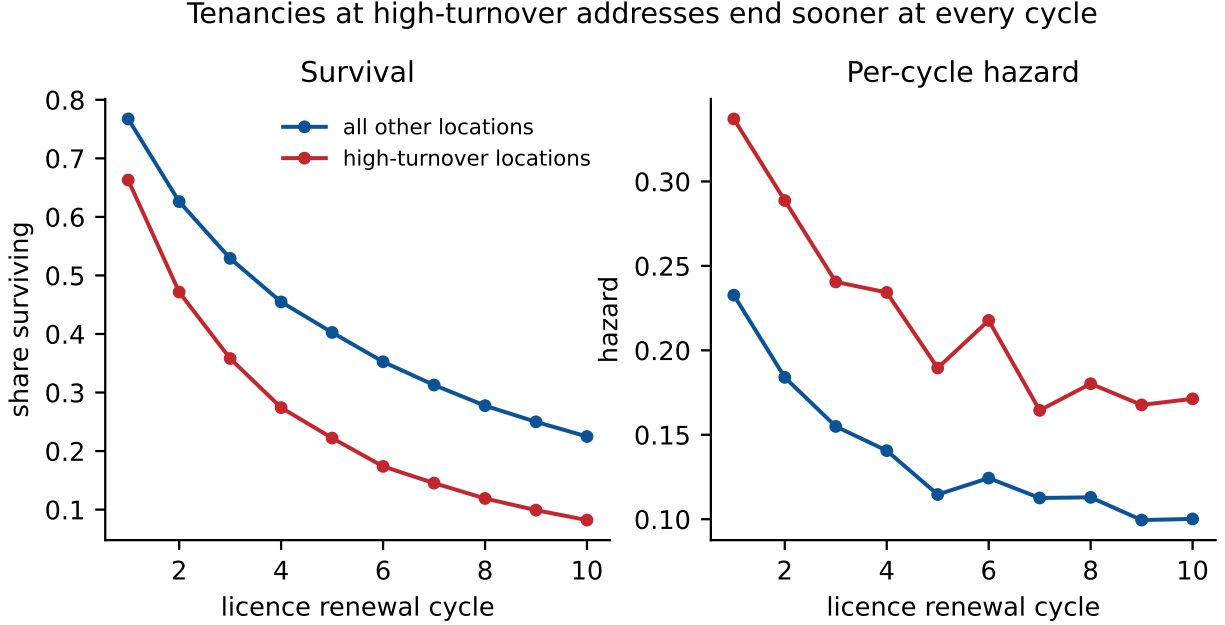


Figure 5: Discrete-time survival and per-cycle hazard over licence renewal cycles. Renewal cycles are used in place of elapsed days because duration is quantized to licence terms.

The single-city scope limits external validity of the urban finding. The evaluation findings depend on the grouped structure of the features rather than on Chicago specifically, but replication across cities remains future work.

Excess turnover is defined relative to a model, so the label inherits that model’s assumptions. Three alternative definitions are reported in the supplement.

8 Conclusion

Spatially blocked cross-validation protects against optimistic evaluation only when blocks are large relative to the spatial support of the features, and the dominant leakage channel in this task is not proximity but group-aggregate encoding over administrative units. Because that mechanism depends on grouping rather than geography, it applies to any target-encoded grouped data. On our task the difference between the default protocol and a correct one is AUC 0.970 versus 0.693, and a reversal of model selection.

Reproducibility

All inputs are public domain. Code, the download utility with row-count verification, and the derived tenancy and location tables are released at <https://github.com/naazzarov/storefront-turnover-leakage>. The pipeline is deterministic given the seed reported in the repository.

References

- T. Dunne, M. J. Roberts, and L. Samuelson. Patterns of firm entry and exit in U.S. manufacturing industries. *RAND Journal of Economics*, 19(4):495–515, 1988.
- S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 2023.
- S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM TKDD*, 6(4):1–21, 2012.
- D. Micci-Barreca. A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. *ACM SIGKDD Explorations*, 3(1):27–32, 2001.
- P. Ploton et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11:4540, 2020.
- L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin. CatBoost: unbiased boosting with categorical features. In *NeurIPS*, 2018.
- D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.
- R. Valavi, J. Elith, J. J. Lahoz-Monfort, and G. Guillera-Arroita. blockCV: An R package for generating spatially or environmentally separated folds. *Methods in Ecology and Evolution*, 10(2):225–232, 2019.